

CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

AIM :

To generate an Amplitude Modulated (AM) signal in MATLAB using specified carrier and modulating signal parameters

PROGRAM (MATLAB CODE):

% Experiment 2: AM sideband frequencies and power

clear; clc; close all;

% Parameters

Ac = 1; % Carrier amplitude

Am = 0.5; % Message amplitude

fc = 1000; % Carrier frequency (Hz)

fm = 100; % Message frequency (Hz)

mu = Am / Ac; % Modulation index

Fs = 20000; % Sampling frequency (Hz)

T = 0.1; % Signal duration (s)

t = 0:1/Fs:T-1/Fs;

% Generate AM signal

message = Am * cos(2*pi*fm*t);

carrier = Ac * cos(2*pi*fc*t);

```
amSignal = (Ac + message) .* cos(2*pi*fc*t);
```

```
% Plot time-domain signals
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(t, message);
```

```
title('Message Signal');
```

```
xlabel('Time (s)'); ylabel('Amplitude'); grid on;
```

```
subplot(3,1,2);
```

```
plot(t, carrier);
```

```
title('Carrier Signal');
```

```
xlabel('Time (s)'); ylabel('Amplitude'); grid on;
```

```
subplot(3,1,3);
```

```
plot(t, amSignal);
```

```
title(['AM Signal, modulation index \mu = ', num2str(mu)]);
```

```
xlabel('Time (s)'); ylabel('Amplitude'); grid on;
```

```
% Calculate and plot the single-sided spectrum
```

```
N = length(amSignal);
```

```
Y = fft(amSignal);
```

```
P2 = abs(Y/N);
```

```
P1 = P2(1:floor(N/2)+1);
```

```
P1(2:end-1) = 2*P1(2:end-1);
```

```
f = Fs*(0:floor(N/2))/N;
```

```

figure;
plot(f, P1);
xlim([0 2*fc]);
title('Single-Sided Amplitude Spectrum of AM Signal');
xlabel('Frequency (Hz)'); ylabel('Amplitude'); grid on;

% Theoretical sideband frequencies and powers
fcLower = fc - fm;
fcUpper = fc + fm;
Pc = Ac^2/2;           % Carrier power for 1-ohm load
PsbEach = Pc * mu^2 / 4; % Power in each sideband
PsbTotal = 2 * PsbEach;
Ptotal = Pc + PsbTotal;

fprintf('Lower sideband frequency: %.2f Hz\n', fcLower);
fprintf('Carrier frequency: %.2f Hz\n', fc);
fprintf('Upper sideband frequency: %.2f Hz\n', fcUpper);
fprintf('Carrier power: %.4f W\n', Pc);
fprintf('Power in each sideband: %.4f W\n', PsbEach);
fprintf('Total sideband power: %.4f W\n', PsbTotal);
fprintf('Total transmitted power: %.4f W\n', Ptotal);

```

OUTPUT:



